

How fast does Normal happen?

Shayaan Tanveer · Math 167R · R + Quarto

There is no universal $n = 30$. I investigated the smallest tested sample size I can defend as approximately Normal across five required populations and three extra-credit processes.

My decisions, and the argument behind them

Case	n	Why I accept it	Why not smaller?
Normal	1	Exact Normality at every n.	No positive n below 1.
Six-sided die	10	Symmetric bell shape; modest short-tail error and fine steps.	5 has clearer short-tail error; 6-9 are borderline.
Exponential	90	Good central shape; modest right skew, correct SE.	30-60 bend more; 75-80 are borderline across seeds.
Binomial small	75	Modest skew; mostly lattice steps, correct SE.	20-50 show stronger steps/skew; 60 is borderline.
Cauchy	None	No finite n: averages remain Cauchy.	Larger n cannot fix missing mean and variance.
Dependent failure	350	Bell-shaped empirical means; modest skew. No iid SE claim.	100-200 curve more; 300 is borderline. CLT unproved.
Basketball shots	140	Small skew; fine steps and correct varying-p SE.	30-100 have larger gaps; 110-125 are borderline.
Synthetic ages	15	Irregular mixture smooths quickly; modest left skew.	5-10 are more asymmetric; 12 is borderline.

My standard is practical, not exact. I accept small tail bends and discrete steps when the histogram, Q-Q plot, shape diagnostics, valid theory and nearby larger n jointly support the approximation. This is not a guarantee for extreme-tail probabilities.

Boundary uncertainty: these are judgment calls, not proved transition integers. A borderline smaller n may also be defensible under a slightly looser tolerance. I do not claim a nicer larger plot proves that its neighbor fails.

Center and spread are not shape.

n counts observations inside one sample. $B = 10,000$ counts separate simulated samples. More B makes the simulation more precise; more n changes the distribution of the mean.

I started broad, then refined separately for each population. I compare histograms, Q-Q plots, skewness, excess kurtosis, mean and SE. Q-Q RMSE compares sorted, empirically standardized means with standard-Normal quantiles. It is a diagnostic, not a pass/fail test.

Calibration without a fake cutoff. Thirty repeated Normal experiments at each coarse n show ordinary Monte Carlo noise. At $n = 1$, the calibration 2.5th-97.5th percentile range for Q-Q RMSE is 0.010-0.018. Other populations need not match that noise to be useful approximations.

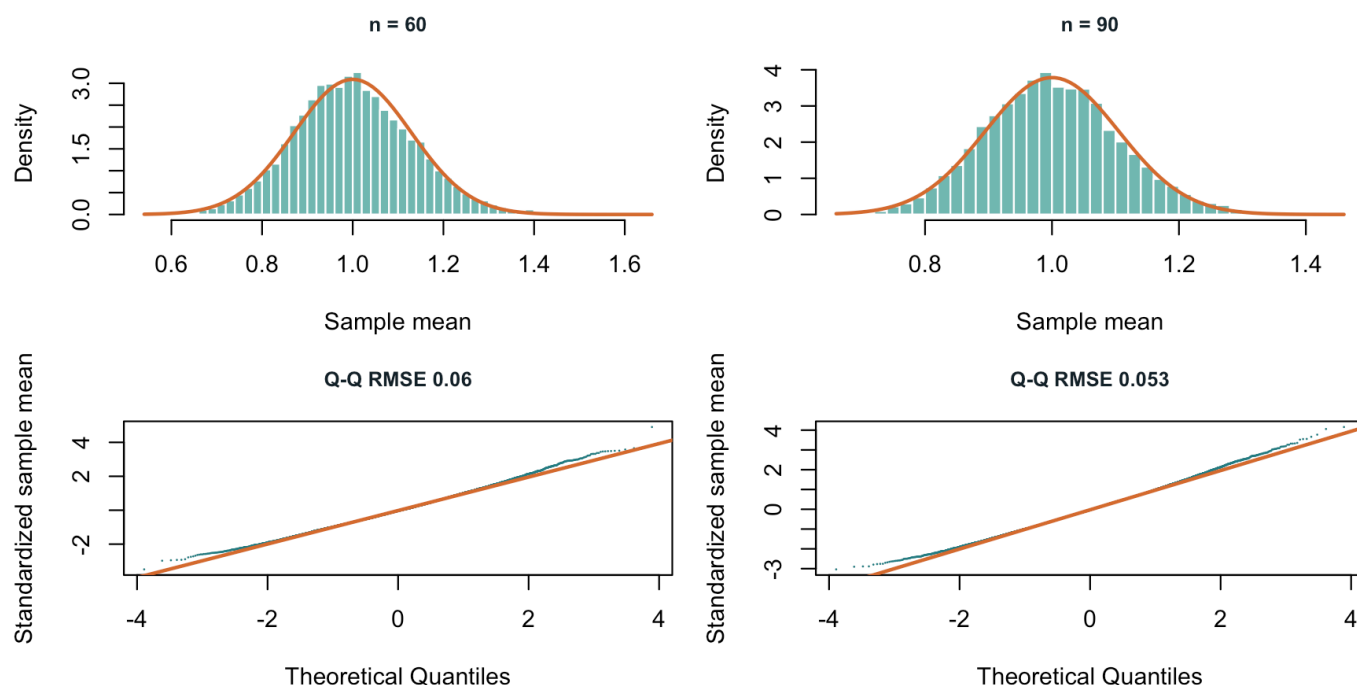


Figure 1: Exponential boundary evidence. Seed 7616, $B = 10,000$. Orange histogram overlays use mean 1 and $SE\ 1/\sqrt{n}$. Residual Q-Q curvature is visible even at the accepted n .

Why 90 rather than 60? The tail bend and skew are smaller, both seeds support similar discrepancy, and larger nearby values remain comparable. The theoretical skewness at 90 is 0.211, not zero. Values 75-80 are borderline, not proven failures.

Slow is not the same as impossible.

Cauchy: increasing n cannot rescue the mean

Independent standard Cauchy averages remain standard Cauchy at every n . Population mean and variance do not exist; $\sigma/\sqrt{n}$ is invalid. Finite simulated means and SDs are calculable, but do not estimate existing population moments. Testing through 5,000 illustrates this theoretical failure.

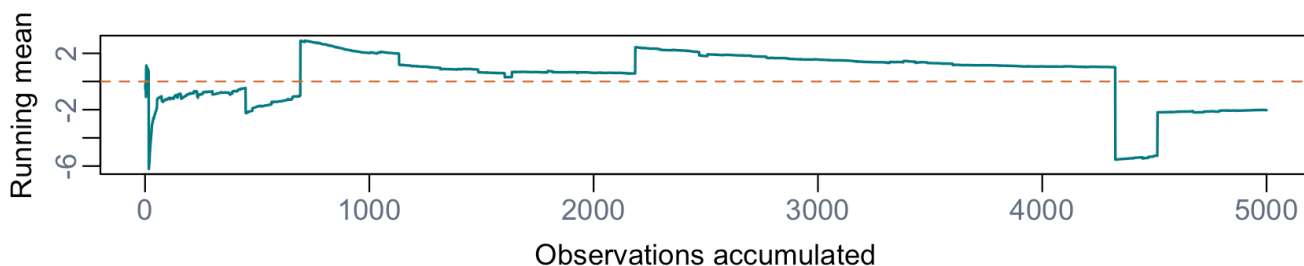


Figure 2: One Cauchy sequence, seed 7616. A running mean remains sensitive to extreme observations; the zero line marks location, not a population mean.

Dependence: a qualified empirical approximation

The next failure rate is $\exp(1.25 \times (\text{previous time} - 1))$. I preserve that recurrence within every sample. A long time tends to be followed by a shorter time. Covariance enters $\text{Var}(\bar{X})$, so I report the **empirical SE**, not an iid theoretical SE.

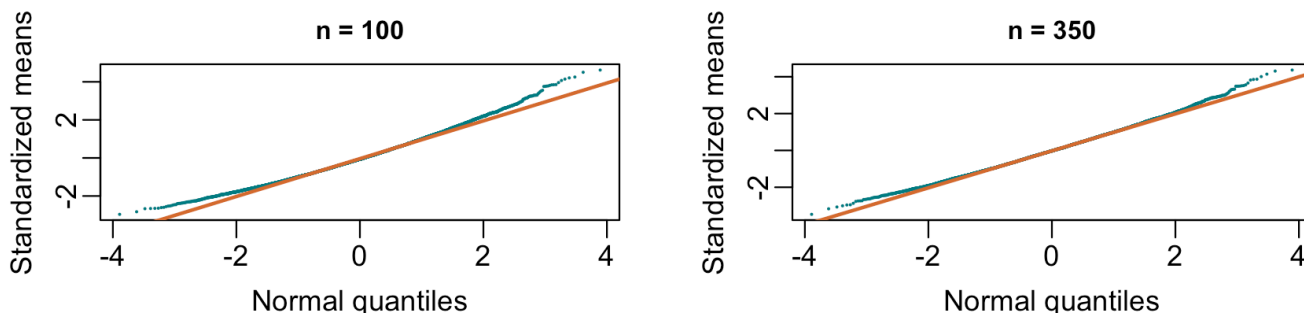


Figure 3: Dependent failure, seed 7616. At 350 the central Q-Q shape is closer to linear, but residual right skew remains. This is not a proof of a dependent CLT.

Individual times have bounded second moments because rates are bounded below. That does **not** establish a dependent CLT. My choice of 350 is empirical and qualified; exact boundary and asymptotic behavior remain unresolved. This differs fundamentally from Cauchy's proved non-convergence to Normal.

The numbers behind the choices.

All values below come from generated diagnostic rows at the selected n . **Kurtosis means excess kurtosis.** Q-Q values use seeds 7616 and 7617; N/A indicates an invalid or unavailable quantity, not zero.

Case	n	Skew	Kurtosis	QQ 7616	QQ 7617	SE sim.	SE theory
Normal	1	0.003	-0.022	0.011	0.011	1.001	1.000
Six-sided die	10	0.000	-0.146	0.057	0.058	0.539	0.540
Exponential	90	0.214	0.032	0.053	0.054	0.106	0.105
Binomial small	75	0.109	0.050	0.057	0.060	0.077	0.077
Cauchy	None	N/A	N/A	N/A	N/A	N/A	N/A
Dependent failure	350	0.208	0.097	0.051	0.049	0.115	N/A
Basketball shots	140	-0.052	-0.039	0.057	0.061	0.038	0.037
Synthetic ages	15	-0.191	-0.039	0.050	0.056	7.128	7.135

Seed sensitivity: exact diagnostics change, but the selected values remain practical candidates under both seeds. The exact minimum is not immutable. I use one fixed alternate seed, never whichever seed produces the nicest picture. Saved vectors and CSV rows are cross-checked before plotting.

What mattered beyond the ordinary iid CLT

Basketball: independent shots have n probabilities spanning .90 to .50 ($n = 1$ uses .90). Expected mean is their average; variance is $\sum[p_i(1-p_i)]/n^2$. The iid $\bar{p}$ shortcut overstates variance. Independence is preserved; identical distribution is not needed for this bounded independent case.

Synthetic ages: the instructor's irregular mixture on 0-100 has mean 49.981 and SD 27.635. It is not historical mortality data. A complex-looking population need not be the slowest to produce a useful Normal approximation.

AI iteration, and my responsibility

I prompted AI to build and revise the R project. The revision fixed overwritten seed outputs, inappropriate shared refinement grids and overly strict benchmark matching. I questioned whether choosing a nicer plot answered the smallest- n question; that led to smaller-candidate checks and explicit boundary uncertainty. I approved a practical standard, not exact tail accuracy.

My role is prompting, checking the evidence and understanding the important code. The interaction log records actual exchanges; it does not claim a personal pre-run prediction or complete line-by-line mastery. Each separate investigation includes theory, plots, diagnostics, decision reasoning and code explanations.

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